MILITARY SPECIFICATION

IBUPROFEN TABLETS

This specification is approved for use by the Defense Personnel Support Center, Defense Logistics Agency, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Ibuprofen Tablets.

1.2 Classification. The tablets covered by this specification shall be of the following sizes (see 6.2):

Size 1 - 60s
Size 2 - 500s
Size 3 - individually sealed, 100s

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on the date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals and Pharmaceuticals.

STANDARDS

FEDERAL

FED-STD-140 - Tablets (for Medicinal Purposes).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. DEPARTMENT OF HEALTH, EDUCATION AND WELFARE,
FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia (USP).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

The National Formulary (NF).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be Ibuprofen Tablets, and shall be in accordance with all applicable requirements of FED-STD-140, and as specified herein.

S2. Classification. Shall be type I, class 2.

S5.3 The following is added to this paragrpah:

Shall be coated tablets containing 400 mg of ibuprofen in each tablet, within the designated assay limits for the tablets.

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S6.4.9 Disintegration. The tablets shall meet the requirements of the USP Disintegration Test for Uncoated Tablets, in not more than 30 minutes.

S6.4.11 Moisture Content. Shall not exceed 5.0 percent when determined by the USP Titrimetric (Karl Fischer) method, using at least 5 ground tablets.

3.1.1 Assay. The tablets shall assay to contain not less than 90.0 percent and not more than 110.0 percent of the required amount of ibuprofen, when assayed as specified in 4.5.1.

3.1.2 Identification. The ibuprofen in the tablets shall be identified when tested as specified in 4.5.2.

3.1.3 Dissolution of the tablets. Not less than 50 percent of the required amount of ibuprofen shall be in solution after 45 minutes when tested as specified in 4.5.3.

3.2 Ingredients.

3.2.1 Ibuprofen.

3.2.1.1 Water content. Shall have not more than 1.0 percent when tested using the USP Water Determination Titrimetric Method.

3.2.1.2 Residue on ignition. Shall have not more than 0.5 percent when determined as specified in 4.5.4.

3.2.1.3 Heavy metals. Shall have not more than 20 ppm using the USP method for heavy metals, Method II.

3.2.1.4 Identification. The elution pattern of the sample compares qualitatively to that of the Ibuprofen Reference Standard preparation when tested under identical conditions of vapor phase chromatography.

3.2.1.5 Assay. Shall assay to contain not less than 97.0 percent and not more than 103.0 percent of ibuprofen calculated on the anhydrous basis when assayed as specified in 4.5.5.

3.3 Other ingredients. All other ingredients used in the manufacture of the tablets shall comply with S5.1 of FED-STD-140. As an alternate to USP or NF testing of the inactive ingredients entering into the finished product, the inactive ingredients shall be tested according to the procedures and comply with the limits in the firm's New Drug Application.

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3.4 Expiration dating period. The expiration dating period shall be not less than 18 months.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own ot any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests preformed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the item to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Sampling. Sampling as shown in FED-STD-140 shall be applicable.

4.3 Packaging inspection. The inspection of the intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Labeling inspection. Sampling shall be in accordance with MIL-STD-105 and the unit of product shall be the unit of issue. The inspection lot shall be the number of units of issue of each batch, lot, control number, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

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4.5 Tests.

4.5.1 Assay for Ibuprofen (finished tablets).

Instrument and Column Conditions:

Instrument and column may vary provided that equivalent chromatography and results are produced.

Instrument:	Hewlett-Packard 402 or comparable instrument
Column:	120 cm x 3 mm I.D. glass
Column Packing:	10 percent silicone oil DC-710 on 80-100 mesh Diataport S or Gas Chrom Q
Column Temperature:	185°C (Isothermal)
Helium:	Approximately 60 ml/min.
Sample size:	0.5 to 2.0 microliters

Column Conditioning: Condition for at least 2 hours at 255°C with carrier gas flow at 60 ml/min. Inject 5 µl portions of Silyl-8 on the column during this period. Turn the temperature down to the assay temperature and, if necessary, inject addtional aliquots of Silyl-8 until a stable baseline is achieved.

Reagents:

Chloroform
Nonadecane (n-19)
Hexamethyldisilazane
Anhydrous Sodium Sulfate

Internal Standard Preparation:

Prepare a chloroform solution containing approximately 2.2 mg of nonadecane (n-19) per ml.

Standard Preparation:

Weigh accurately approximately 20 mg of Ibuprofen Reference Standard and transfer to a suitable flask and add 10.0 ml of Internal Standard Preparation. Add 1 ml of hexamethyldisilazane to the solution, warm to 40-45°C, shake vigorously, and allow to stand for 30 minutes.

Sample Preparation:

Place sufficient tablets, equivalent to about 1000 mg of ibuprofen, into a 250 ml separator. Add 25 ml of 0.01 N hydrochloric acid and 200.0 ml of chloroform; shake vigorously for 60 minutes. Filter about 10 ml of the chloroform layer into a suitable vial, which contains about one gram of anhydrous sodium sulfate, stopper the vial and shake vigorously.

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Pipet an accurately measured aliquot of the dried chloroform extract equivalent to 20 mg of Ibuprofen into a vial, add 10.0 ml of the Internal Standard Preparation and mix. Add 1 ml of hexamethyldisilazane to the solution, warm to 40°-45°C, shake vigorously, and allow to stand for 30 minutes.

Procedure:

Use the aliquot quantity of the Sample Preparation and the Standard Preparation according to the described conditions.

Calculations:

$$\text{mg Ibuprofen per tablet} = \frac{R(sa)}{R(std)} \times \frac{Cs}{WT(sa)} \times \frac{F1}{F3} \times F2 \times \text{Ave Tab WT (mg)}$$

where,

$R(sa)$ = Area of the ibuprofen sample peak / Area of the internal standard peak

$R(std)$ = Area of the ibuprofen standard peak / Area of the internal standard peak

Cs = Weight of Ibuprofen Reference Standard in mg

$WT(sa)$ = Weight of the sample in mg

$F1$ = ml of Internal Standard Preparation used in the Sample Preparation.

$F2$ = Dilution factor, 200 ml + aliquot taken.

$F3$ = ml of Internal Standard Preparation used in the Standard Preparation.

4.5.2 Identification. Grind one tablet to a fine powder in a mortar, add approximately 5 ml of chloroform to the powder and swirl to dissolve the Ibuprofen. Filter the mixture through filter paper and evaporate the chloroform to dryness with a stream of air or nitrogen. Prepare a mineral oil mull of the residue and measure its infrared absorption spectrum in the range of 2 to 15 microns: the spectrum shall exhibit maxima only at the same wavelengths as that of a mineral oil mull of Ibuprofen Reference treated in the same manner.

4.5.3 Dissolution of the tablets. Select 6 tablets at random and determine their individual dissolution rates using the USP method except use USP phosphate buffer, pH 7.2, as the dissolution fluid in which the basket is rotated at 150 rpm. Determine the absorbance of filtered portions of the dissolution fluid at a wavelength of about 265nm after 45 minutes comparing it with a solution of known concentration of Ibuprofen Reference. All 6 tablets should meet the requirement of at least 50 percent dissolved in 45 minutes. If 1 or 2 tablets fail to meet the requirement, repeat the test on 6 additional tablets: not less than 10 of the 12 tablets must meet the requirement.

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4.5.4 Residue on ignition. Place approximately 1 gram of the sample, accurately weighed, in a tared porcelain crucible and carefully ignite at a low temperature until thoroughly charred. The crucible may be loosely covered with a porcelain lid during the charring. Add to the contents of the crucible 2 ml of nitric acid and 5 drops of sulfuric acid, and cautiously heat until the white fumes are evolved, then ignite, preferably in a muffle furnace, at 500°C to 600°C until the carbon is all burned off. Cool the crucible in a desiccator and weigh. From the weight of the residue obtained, calculate the ash content.

4.5.5. Assay for ibuprofen (ingredient). Use the vapor phase chromatographic procedure as described for the tablets (see 4.5.1) except for the following changes in the sample preparation and calculation.

Sample Preparation:

Weigh accurately approximately 100 mg of the sample powder and transfer to a suitable flask. Add 50.0 ml of Internal Standard Preparation, 5 ml of hexamethyldisilazane, warm to 40°-45°C, and shake vigorously. Allow the mixture to stand for 30 minutes and continue as directed under Procedure for the tablet assay.

Calculations:

$$\text{percent ibuprofen} = \frac{R(sa)}{R(std)} \times \frac{Cs}{WT(sa)} \times \frac{F1}{F2} \times 100$$

where,

$R(sa)$ = Area of the Ibuprofen sample peak / Area of the internal standard peak

$R(std)$ = Area of the Ibuprofen standard peak / Area of the internal standard peak

Cs = Weight of ibuprofen reference standard in mg

$WT(sa)$ = Weigth of the sample in mg

$F1$ = ml of Internal Standard Preparation used in the Sample Preparation.

$F2$ = ml of Internal Standard Preparation used in the Standard Preparation.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable requirements of PPP-C-186 and as specified herein.

5.2 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.2.1 Unit of issue. One bottle containing 60 tablets or 500 tablets, or one box containing 100 tablets individually sealed, as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Intermediate container.

5.2.2.1.1 Size 1 and size 2. The immediate container for size 1 and size 2 shall be a commercially available container considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other. The closure for size 1 shall be closure P (Child-proof).

5.2.2.1.2 Size 3. Immediate container for size 3 shall be a commercially available strip or blister package considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container shall be altered in any way by each other.

5.2.2.2 Unit package. At the option of the contractor, each unit of size 1 or size 2 may be packaged as specified in 5.2.5 of PPP-C-186. Each unit of size 3 shall be packaged as specified in 5.2.5 of PPP-C-186.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.3.1 Level A. One hundred thirty-two units of issue of size 1 or twenty-five units of issue of size 2 or twenty-four units of issue of size 3 shall be packed as specified in 5.4.4 of PPP-C-186.

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5.3.2 Level B. One hundred thirty-two units of issue of size 1 or twenty-five units of issue of size 2 or twenty-four units of issue of size 3 shall be packed in a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class domestic. Bursting strength of box shall be in accordance with Special Requirements of the applicable table of PPP-B-636. Closure shall be effected in accordance with the appendix of PPP-B-636.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act and shall include the information required below:

5.4.1.1 Immediate container: Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the expiration date.

5.4.1.2 Unit package. Each unit package shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the National Stock Number.

NOTES: (1) The National Stock Number shall not be preceded by any prefix.
(2) Strip labeling may be used.
(3) Label adhesion test shall not apply.
(4) If a unit package is not supplied for sizes 1 and 2, the National Stock Number shall appear on the immediate container label.

5.4.2 Exterior container. Exterior container shall be marked as specified in 5.5.4 of PPP-C-186. Type II shelf life marking as specified in MIL-STD-129 shall also be shown. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE 15°- 30°C (59° - 86°F)."

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6. NOTES

6.1 Ordering data. Procurement documents shall specify the following:

(a) Title, number, and date of this specification.
(b) National Stock Number.
(c) Selection of applicable levels of packaging and packing (see 5.2 and 5.3).

6.2 This specification covers the following item(s):

NATIONAL STOCK NUMBER	ITEM IDENTIFICATION	SPECIFICATION DESIGNATION
6505-00-128-8035	IBUPROFEN TABLETS, 400 mg, 500s	Size 2
6505-01-034-0457	IBUPROFEN TABLETS, 400 mg, 60s	Size 1
6505-01-034-1237	IBUPROFEN TABLETS, 400 mg, Individually sealed, 100s	Size 3

Preparing Activity:
DLA-DM

Project No. 6505-1821

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-I-37676(DLA-DM) IBUPROFEN TABLETS

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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PHILADELPHIA, PA 19101

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